Microprocessors and Microcontrollers

Basic Connection



Needed Equipments

- Microcontroller – ATmega32
- The AVR Programmer / Loader
  - To load your code into the mc

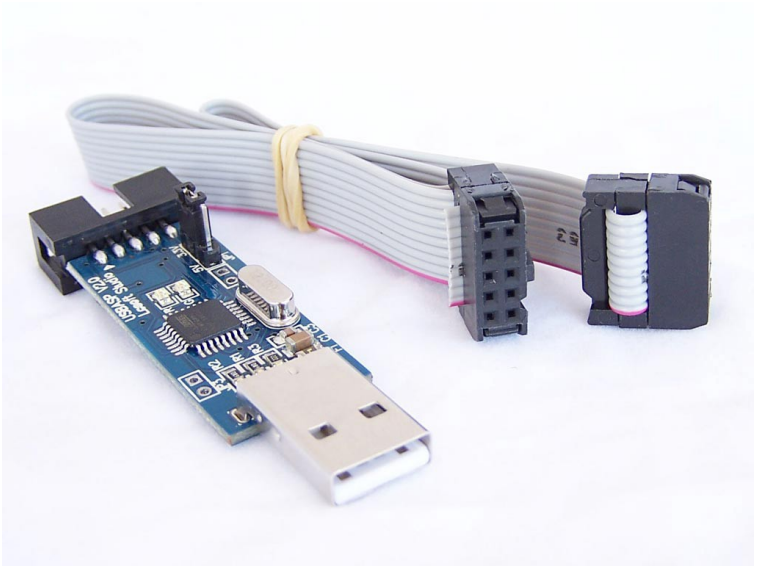


Steps

- Connect USBasp to PC
- (Windows) Check Driver from Device Manager. If problem, update driver using zadig.
- Install Extreme Burner
- Connect USBasp with ATmega32
- Write Code in Atmel
- Upload hex to Atmega32 using Extreme Burner



USBasp Programmer





USBasp Programmer Driver Troubleshoot (only for Windows 10)

Open your web browser and head to https://zadig.akeo.ie/

Using this tool, we first install the correct driver onto the USBasp.

1. Hit the download button and install the application.
2. Open Zadig
3. Click on options and press on *List All Devices*
4. Select USBasp and install the libusbK (v3.0.7.0) driver

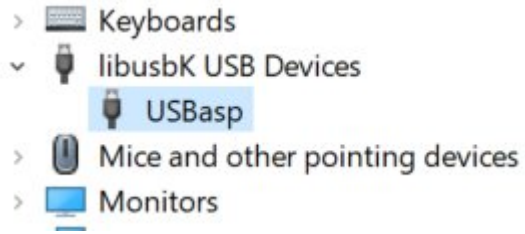
Please note, installing the driver may take up to five minutes.

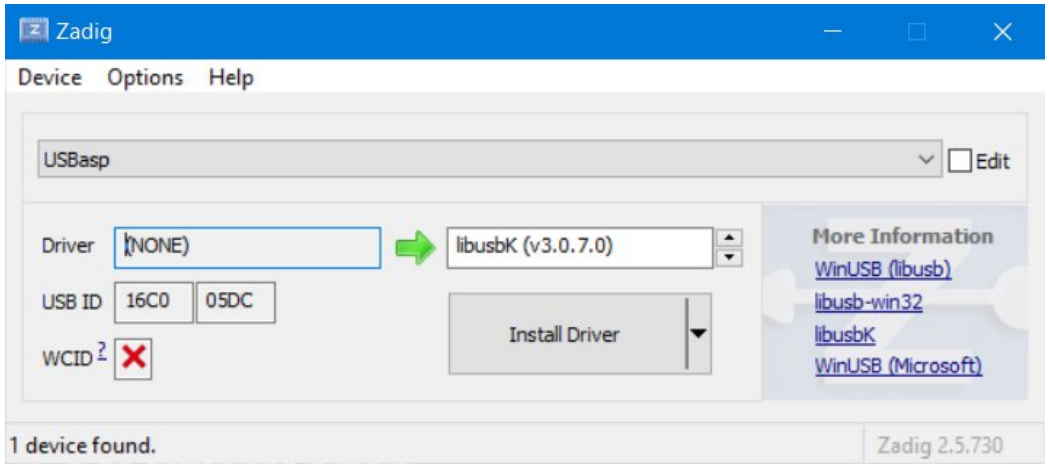
(Step 1: Installing the USBasp Driver Using Zadig) https://www.instructables.com/Programming-Microcontrollers-With-a-USBasp-Program/



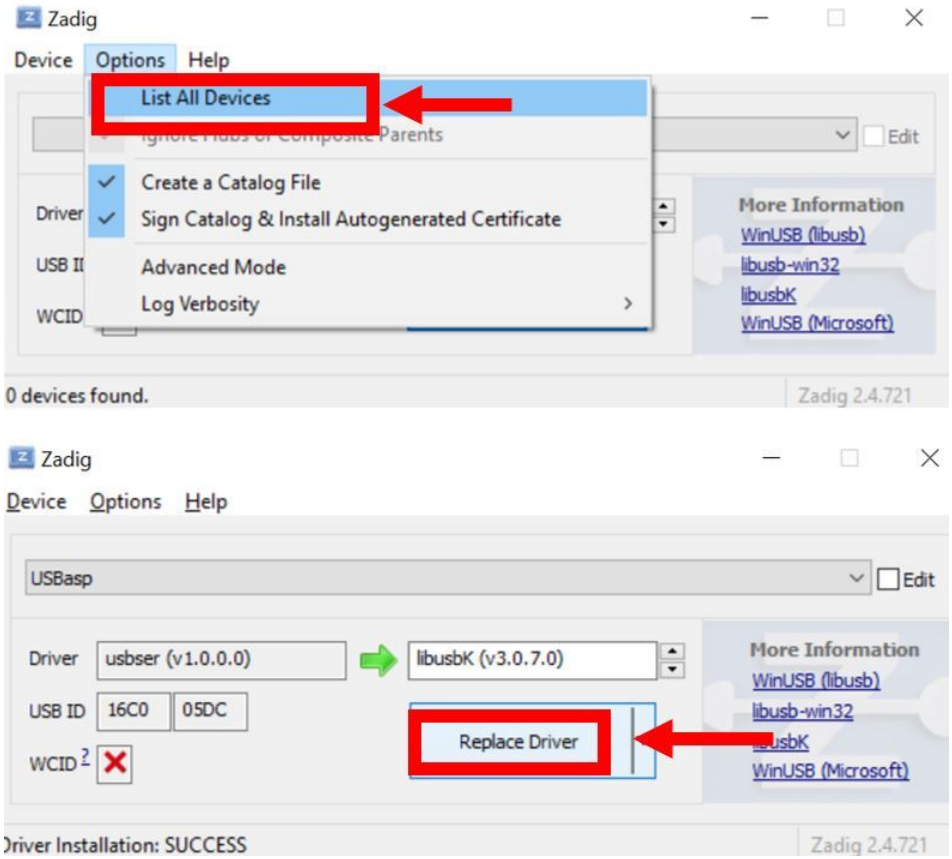
USBasp Programmer Driver Troubleshoot (only for Windows 10)







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(Step 1: Installing the USBasp Driver Using Zadig) https://www.instructables.com/Programming-Microcontrollers-With-a-USBasp-Program/

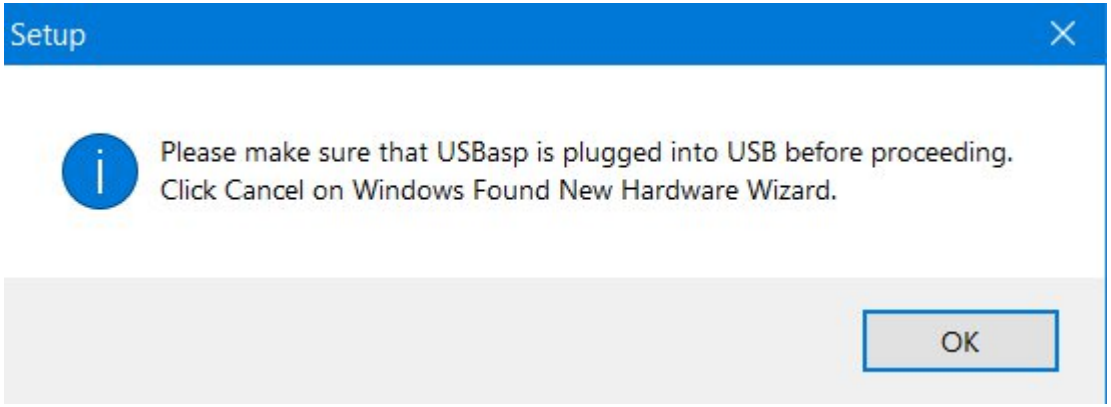
Burner

- Extreme Burner
  - http://extreme-burner-avr.software.informer.com/download/
- There is also avrpal, avrdude

- Loads the HEX file into microcontroller



Burner


Setup
Please make sure that USBasp is plugged into USB before proceeding.
Click Cancel on Windows Found New Hardware Wizard.
OK



Needed Softwares

- AVR Studio
  - https://www.microchip.com/mplab/avr-support/atmel-studio-7
  - **Atmel Studio 7.0 (build 1931) offline installer**

- This is the IDE



Writing Code

- Create New Project
- Select “GCC C Executable Project”
- Select path to save the project
  - You will later need this path for the hex file
- Then select device type
  - Search for ATmega 32
- Now you can write the program



Compile the hex code

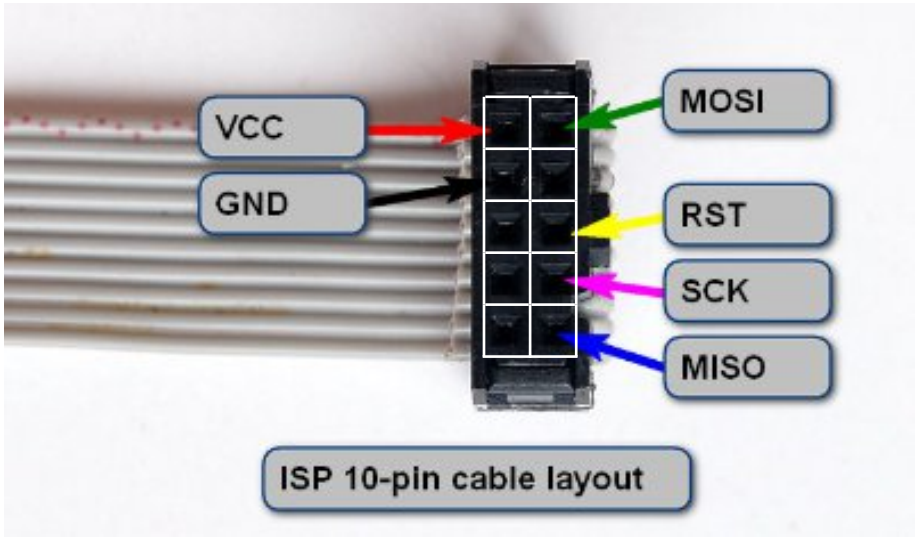
- Finish the program
- Select “Build Solution (F7)” from the Build Menu
- You will find that a hex file is generated in the Debug folder
- The name of the hex file will be the same as your project name



Connecting ATmega32 with programmer

- Following the figure connect the 6 corresponding microcontroller pins with the loader.

- Be careful about the orientation of the cable layout. The extended part goes to the right



Connecting ATmega32 with programmer

1 = VCC (5V)

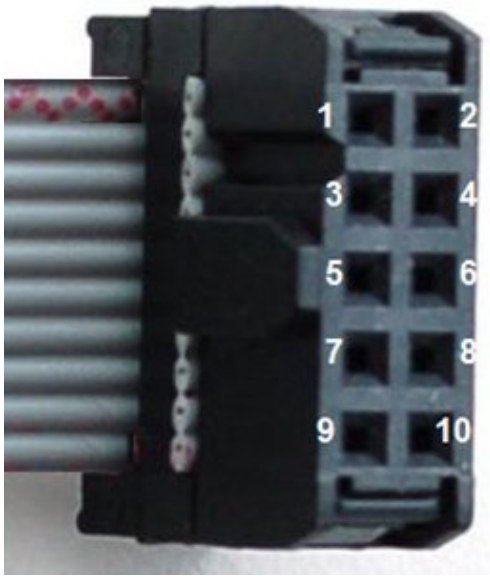
2 = MOSI

3 = GND

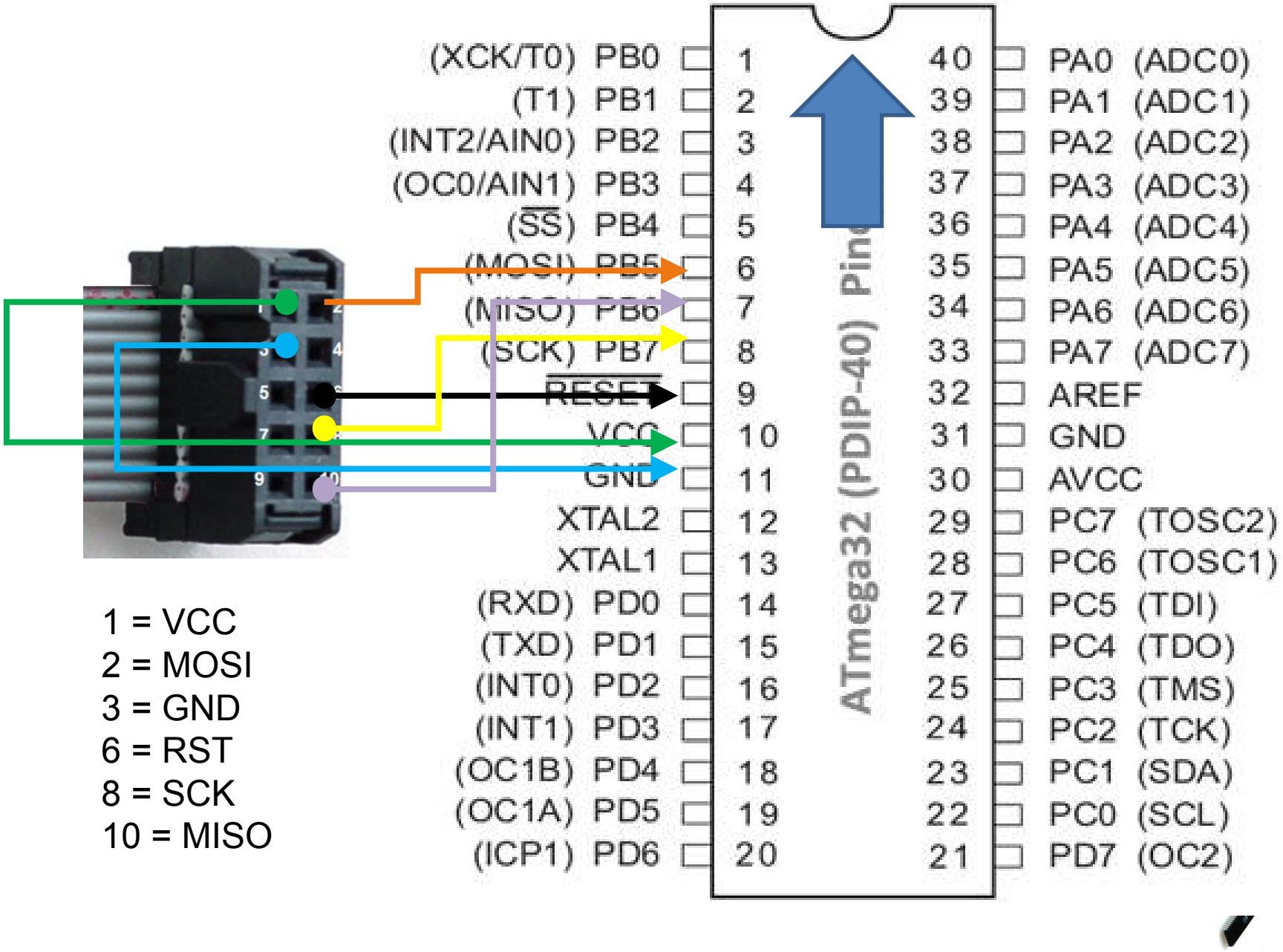
6 = RST

8 = SCK

10 = MISO



Depending on the connection of USBasp programmer, you may have to swap odd pins with even ones (e.g. 1 <-> 2, 3 <-> 4, 5 <-> 6 and so on)

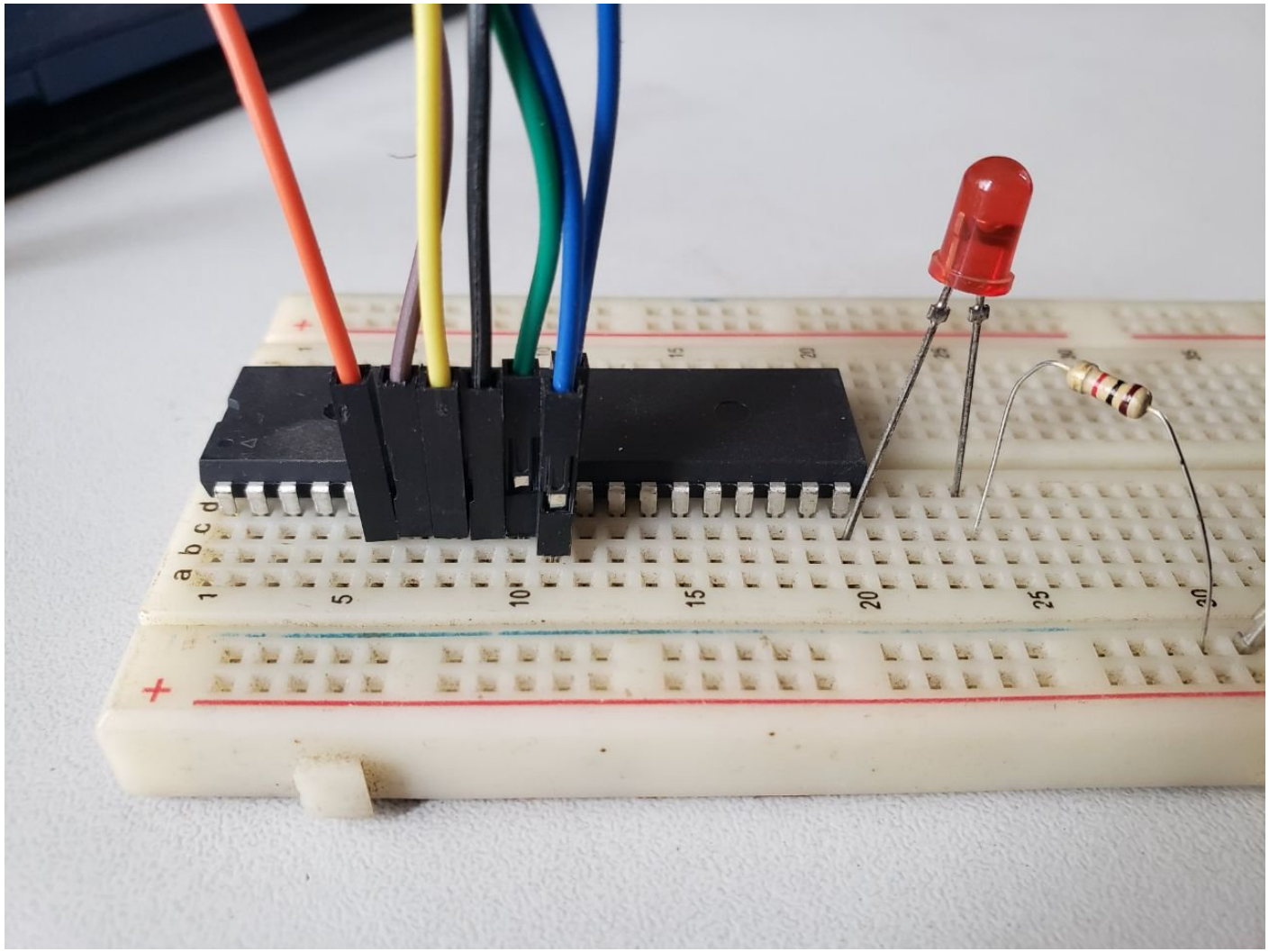

(XCK/T0) PB0 1
(T1) PB1 2
(INT2/AIN0) PB2 3
(OC0/AIN1) PB3 4
(SS) PB4 5
(MOSI) PB5 6
(MISO) PB6 7
(SCK) PB7 8
RESET 9
VCC 10
GND 11
XTAL2 12
XTAL1 13
(RXD) PD0 14
(TXD) PD1 15
(INT0) PD2 16
(INT1) PD3 17
(OC1B) PD4 18
(OC1A) PD5 19
(ICP1) PD6 20
ATmega32 (PDIP-40) Pin
40 PA0 (ADC0)
39 PA1 (ADC1)
38 PA2 (ADC2)
37 PA3 (ADC3)
36 PA4 (ADC4)
35 PA5 (ADC5)
34 PA6 (ADC6)
33 PA7 (ADC7)
32 AREF
31 GND
30 AVCC
29 PC7 (TOSC2)
28 PC6 (TOSC1)
27 PC5 (TDI)
26 PC4 (TDO)
25 PC3 (TMS)
24 PC2 (TCK)
23 PC1 (SDA)
22 PC0 (SCL)
21 PD7 (OC2)
1 = VCC
2 = MOSI
3 = GND
6 = RST
8 = SCK
10 = MISO

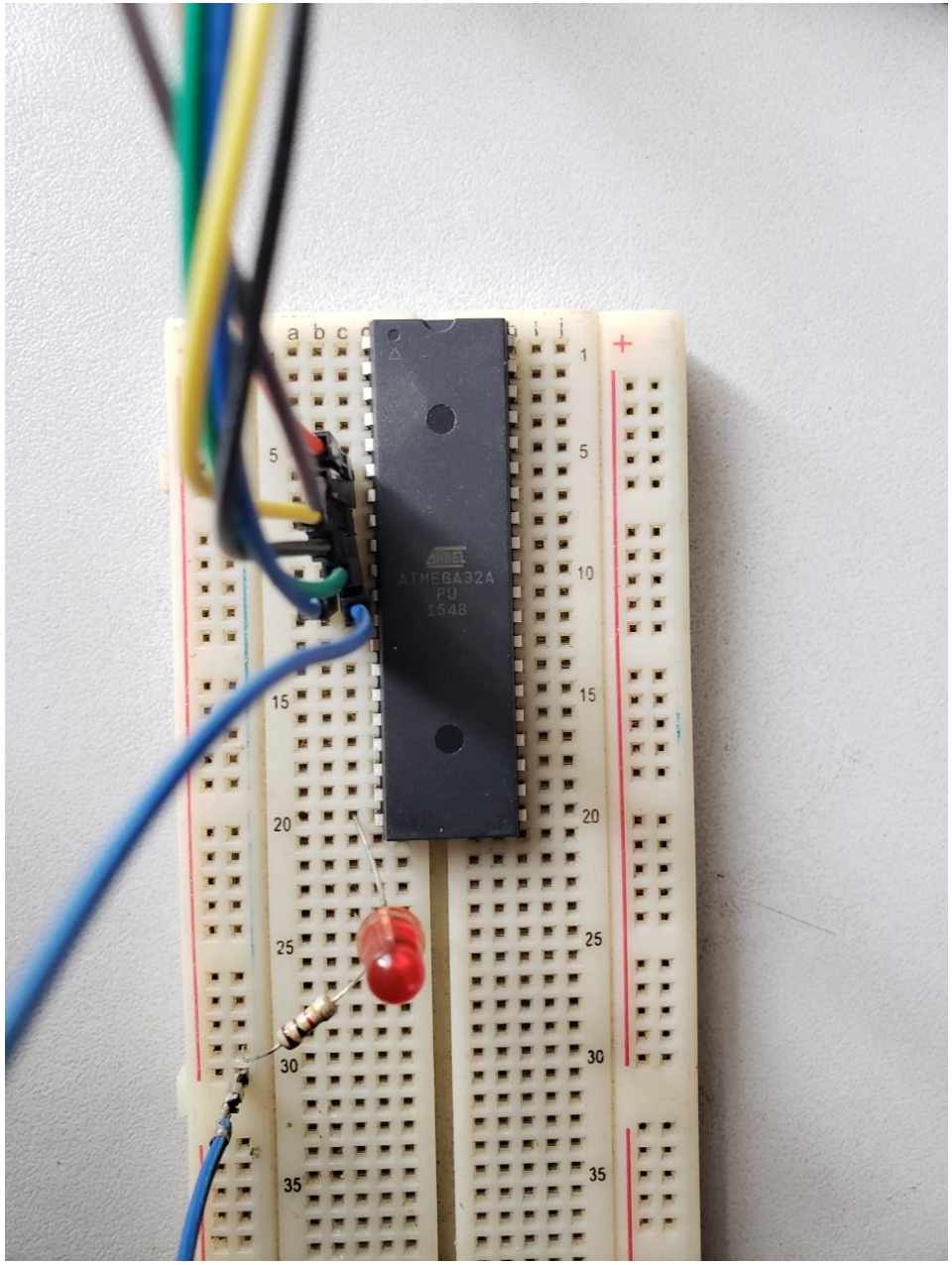
Connecting ATmega32 with programmer

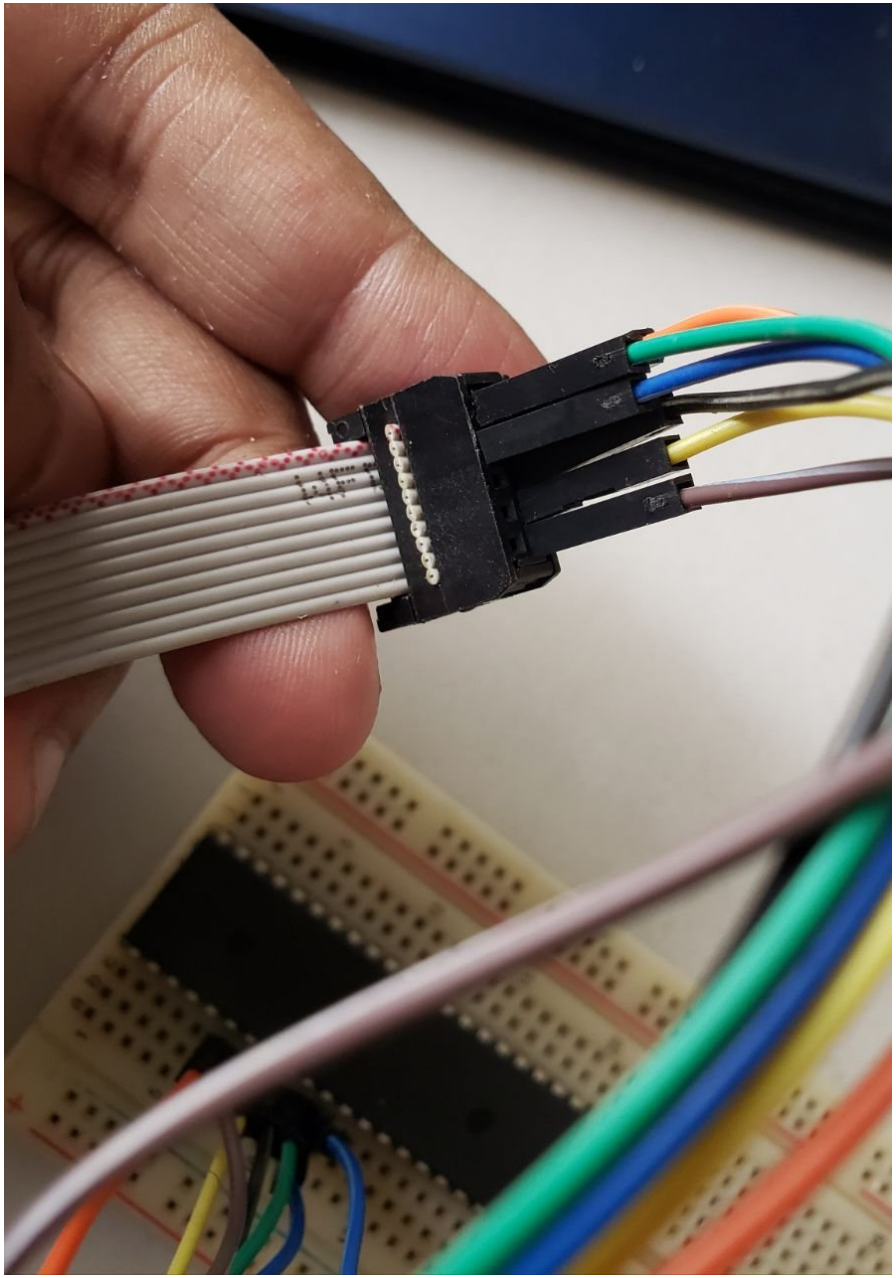


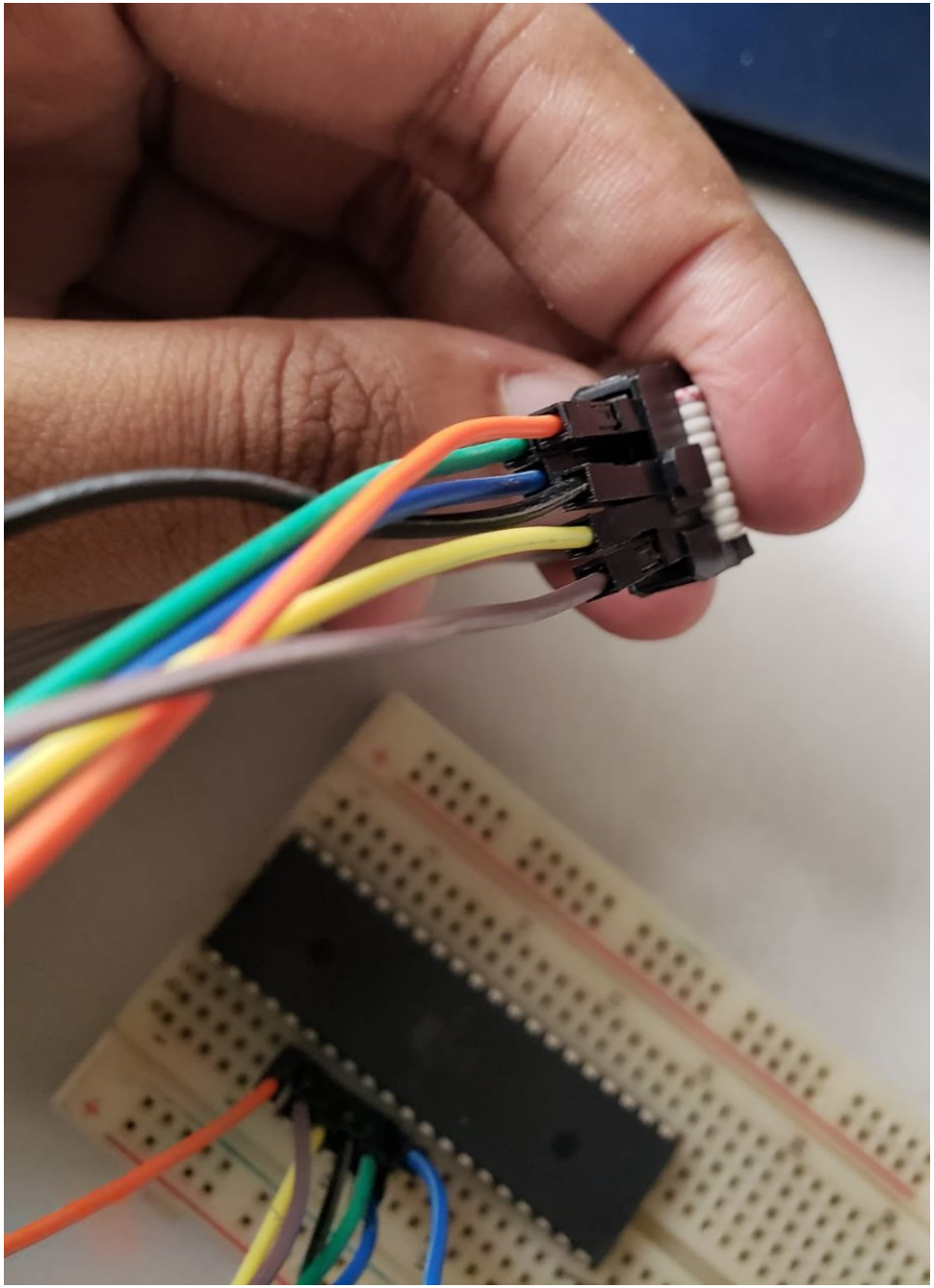
Follow the red wire/stripped wire.
This wire should be #1 pin (VCC).

Today’s Task

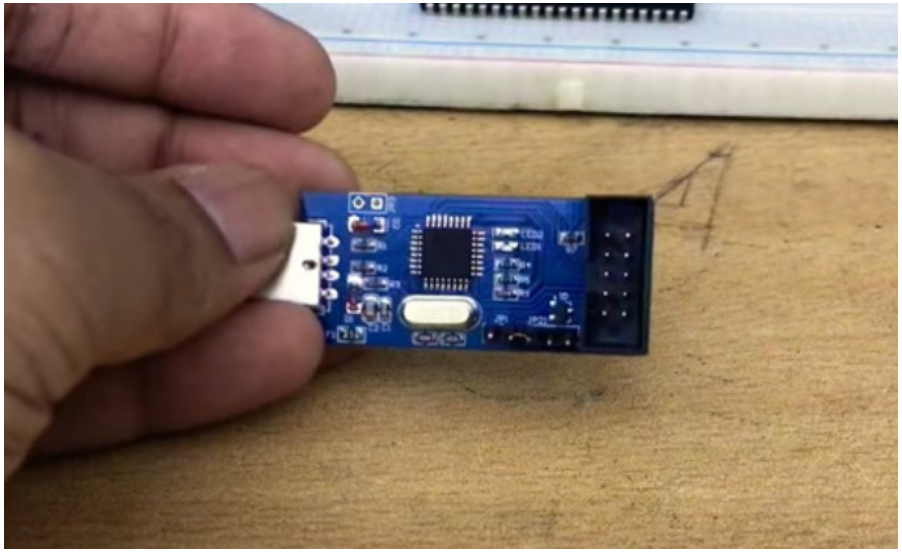


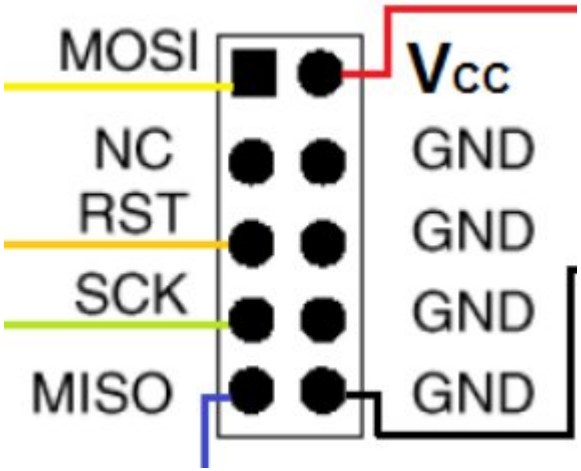

ATMEGA32A
PU
1548





Connecting ATmega32 with programmer (Without Cable)

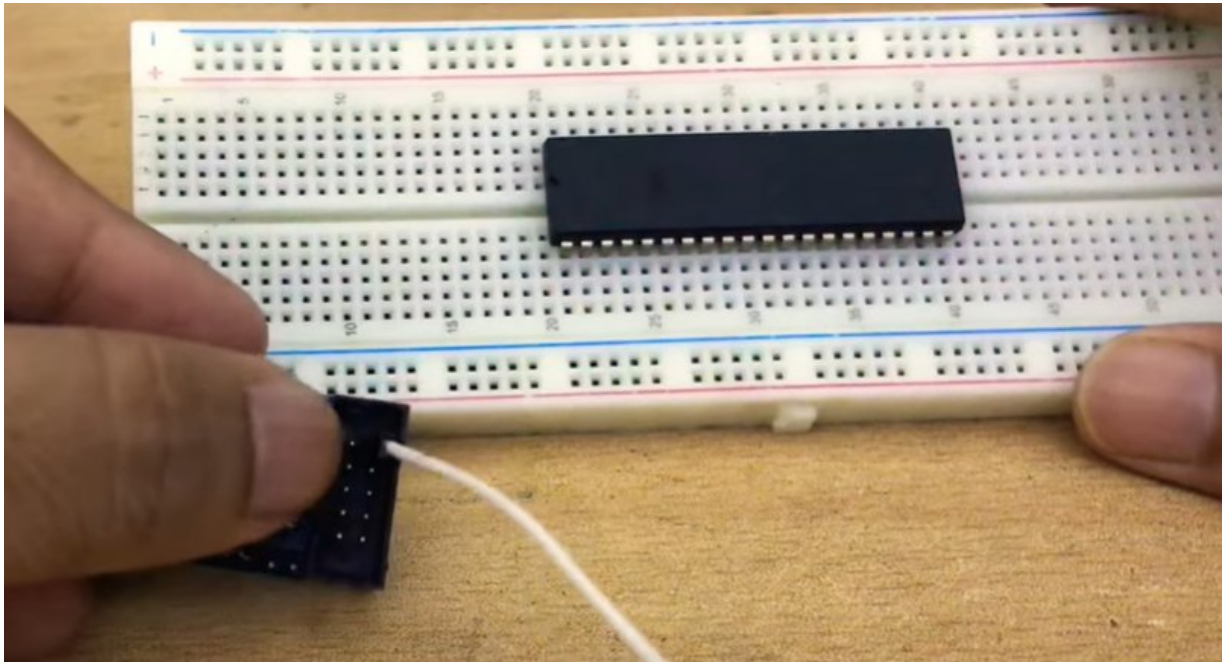




https://www.youtube.com/watch?v=bH8c8sMLfCI

Courtesy: Shanjinur Islam Spondon, CSE 15

Placing ATmega32 on Breadboard

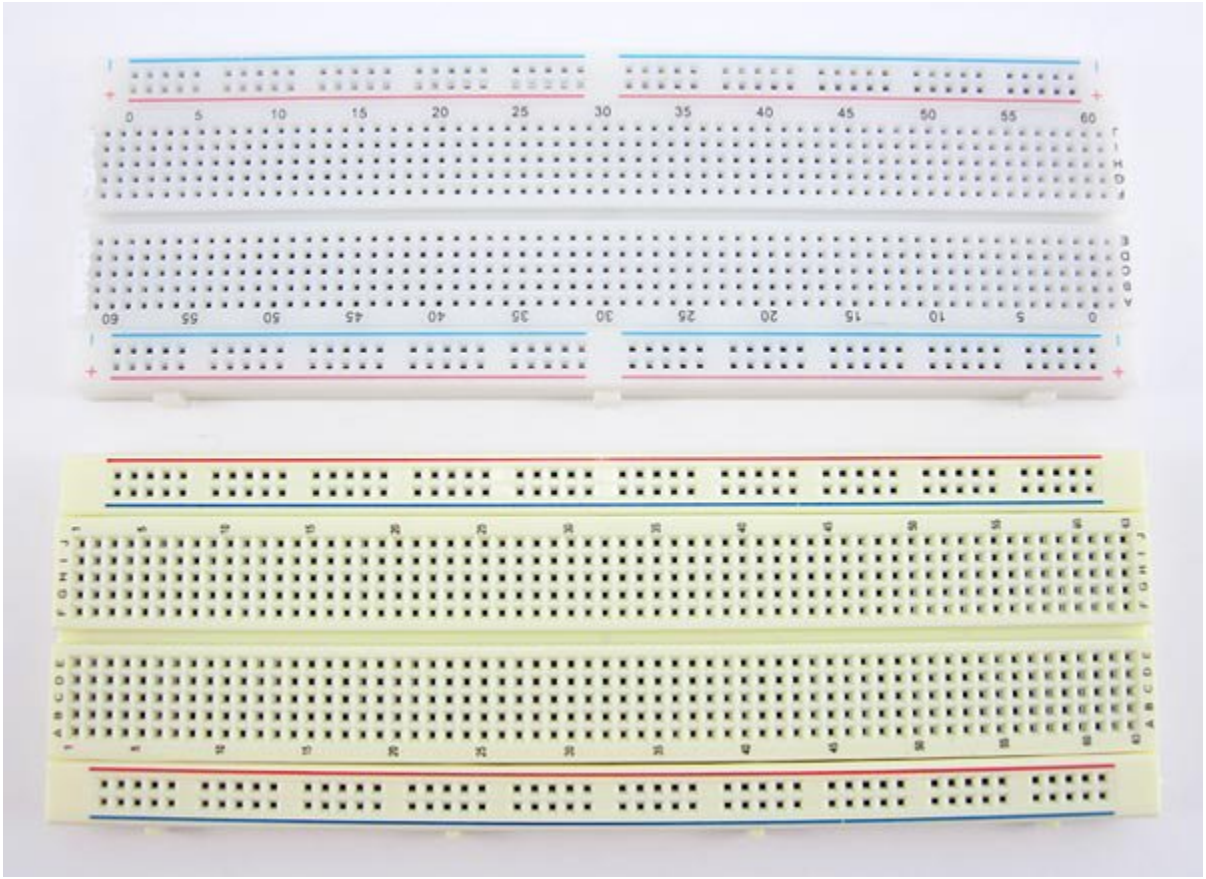


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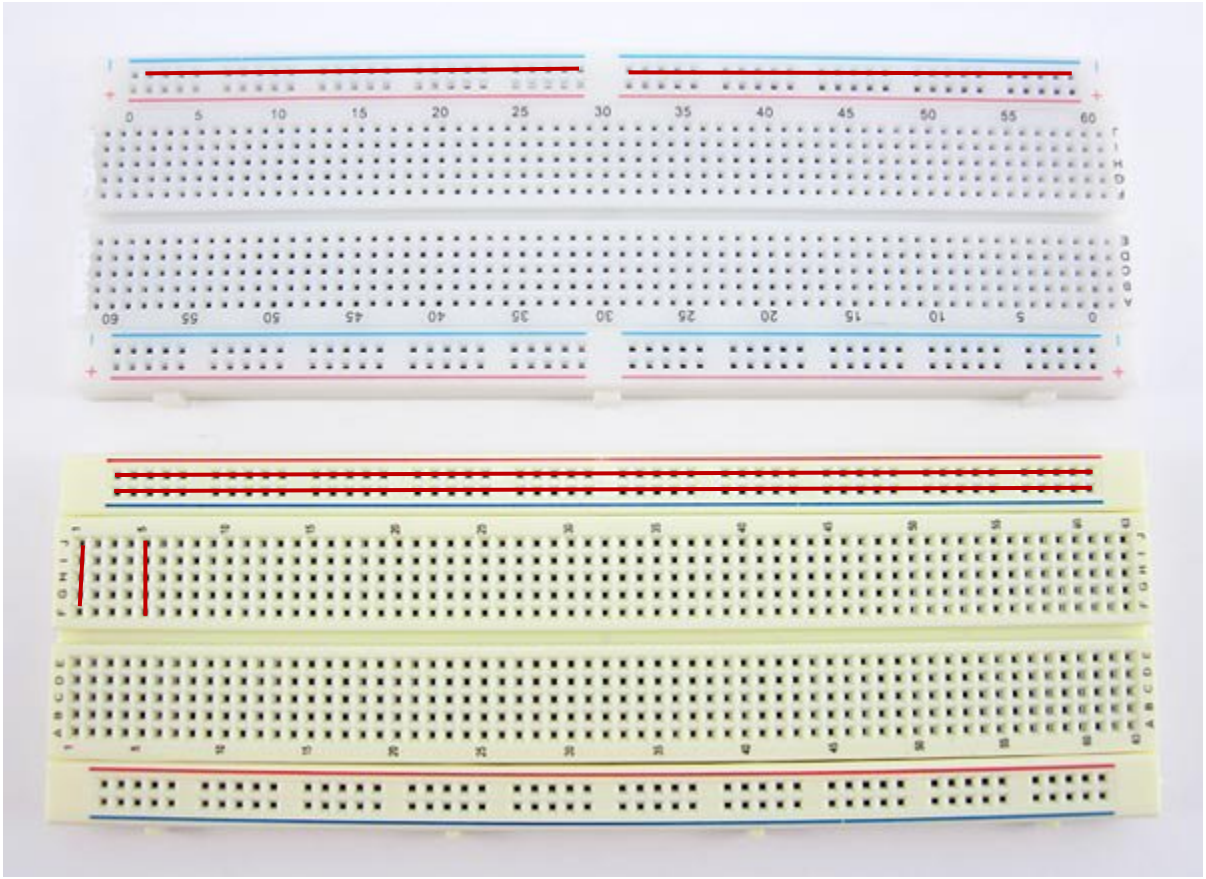
Courtesy: Shanjinur Islam Spondon, CSE 15



Breadboard



Breadboard

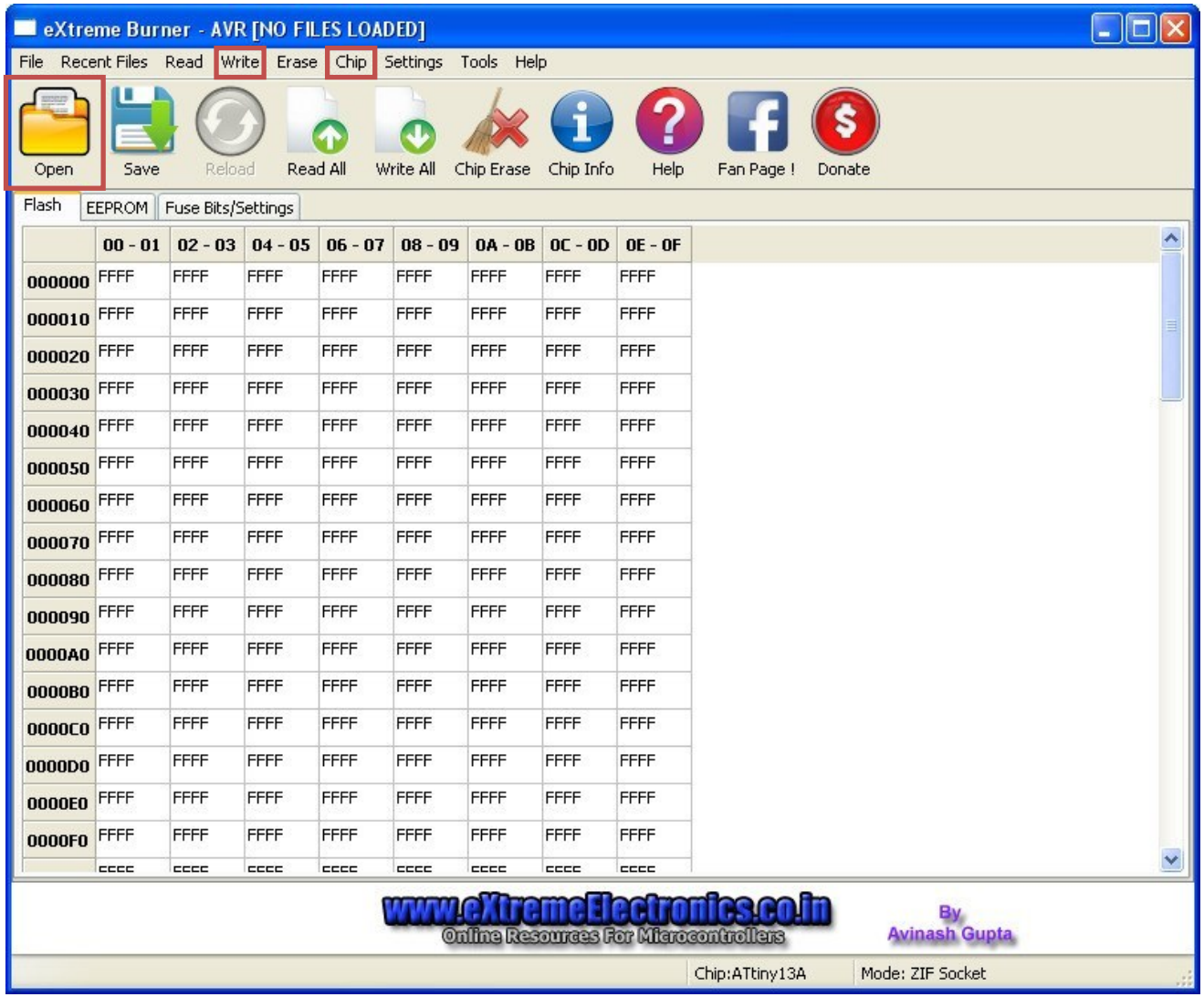


Load the hex into mc

- Run Extreme Burner
- Load the hex file into it by “Open” and locate your hex file
- From menu select Chip -> ATmega32
- Set the microcontroller on the AVR Programmer
- Connect the programmer with PC
- In extreme Burner select the menu “write” and choose “Flash”

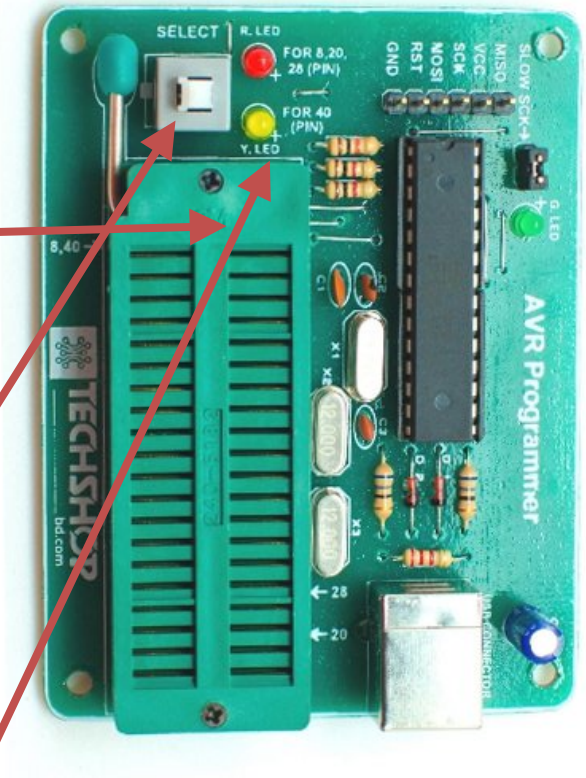


Load the hex into mc



Setting mc in the AVR Programmer (USBasp alternative)

- For the big loaders from TechshopBD
  - The upper portion (U) side goes closer to the side with the liver
  - There is a push button
  - Push it so that the yellow LED is on and the red one is off



Complete Process of Programming a Microcontroller

1. Write the code
   - In ATMEL Studio, or other IDE

2. Compile it to a .hex file.
   - For ATMEL Studio, you will find the hex on the "Debug" folder in your project



Complete Process of Programming a Microcontroller

3. Inserting the .hex file into the microcontroller
   - Load the hex file into burner
   - set the microcontroller on the “AVR Programmer”
   - Write

4. Set the microcontroller in the working circuit



